

6. Method of Characteristics (MOC)

6.1 Introduction

Method of Characteristics (MOC) is a technique of solving partial differential equations (PDEs) by converting them to ordinary differential equations (ODEs) and solving the ODEs by numerical or graphical technique.

Characteristics

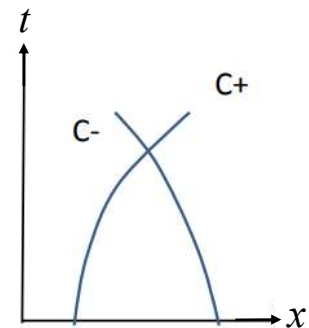
Characteristics are curves for plot of $dx/dt = V \pm c$ in $x-t$ plane, where V = velocity of flow and c = celerity of wave. The term $V \pm c$ represents the velocity of propagation of disturbance.

Slope of curve, $\tan\theta = dt/dx = 1/(V \pm c)$

Slope of characteristic line is $1/(V + c)$ along C^+ and $1/(V - c)$ along C^-

Requirements for solution

- Initial condition: e.g. value of variables at $t = 0$
- Boundary conditions (BC): conditions at the boundary of physical system



Advantages of MOC

- Handling of complex system
- Easy to establish stability of solution
- Easy to handle and program boundary conditions
- Easy to program
- Higher accuracy

Limitation of MOC

➤ Requirements of computer programming for implementation

6.2 Application of MOC to unsteady pipe flow problems

$$\text{Momentum equation: } L_1 = \frac{\partial V}{\partial t} + V \frac{\partial V}{\partial x} + \frac{1}{\rho} \frac{\partial P}{\partial x} + g \sin \theta + \frac{fV|V|}{2D} = 0 \quad (\text{I})$$

$$\text{Continuity equation: } L_2 = \frac{\partial P}{\partial t} + V \frac{\partial P}{\partial x} + \rho c^2 \frac{\partial V}{\partial x} = 0 \quad (\text{II})$$

where V is the average velocity over the section, ρ is the density of fluid, P is the pressure at a point, c is the celerity of wave, g is the acceleration due to gravity, D is the diameter of pipe, $\sin \theta$ is the slope of pipe, and f is the friction factor.

$$c = \sqrt{\frac{K}{\rho(1 + \frac{D K}{E t})}}$$

where K is the Bulk Modulus of fluid; D is the diameter of pipe; E is the Young's modulus of pipe material; and t is the thickness of pipe.

Celerity in pipe flow is constant. The unknown variables to compute are V and P .

Combining two equations as $L = L_1 + \lambda L_2 = 0$. where λ is the multiplier.

$$L = \left[\frac{\partial V}{\partial x} (V + \lambda \rho c^2) + \frac{\partial V}{\partial t} \right] + \lambda \left[\frac{\partial P}{\partial x} \left(\frac{1}{\rho \lambda} + V \right) + \frac{\partial P}{\partial t} \right] + g \sin \theta + \frac{fV|V|}{2D} = 0 \quad (\text{III})$$

We have

$$\frac{dV}{dt} = \frac{\partial V}{\partial x} \frac{dx}{dt} + \frac{\partial V}{\partial t} \text{ and } \frac{dP}{dt} = \frac{\partial P}{\partial x} \frac{dx}{dt} + \frac{\partial P}{\partial t}$$

Substituting in the combined expression

$$L = \frac{dV}{dt} + \lambda \frac{dP}{dt} + g \sin \theta + \frac{fV|V|}{2D} = 0 \quad (\text{IV})$$

The above expression is valid

If $dx/dt = [V + \lambda \rho c^2]$ for velocity part,
and $dx/dt = [1/(\rho \lambda) + V]$ for pressure part

Equating these,

$$V + \lambda \rho c^2 = 1/(\rho \lambda) + V$$

$$\text{i.e., } \lambda = \pm 1/(\rho c)$$

Substituting in equation (IV)

$$L = \frac{dV}{dt} \pm \frac{1}{\rho c} \frac{dP}{dt} + g \sin \theta + \frac{fV|V|}{2D} = 0 \quad (\text{V})$$

With $\lambda = \pm 1/(\rho c)$, dx/dt becomes

$$dx/dt = V \pm c \quad (\text{VI})$$

Equation (V) is valid if equation (VI) is satisfied.

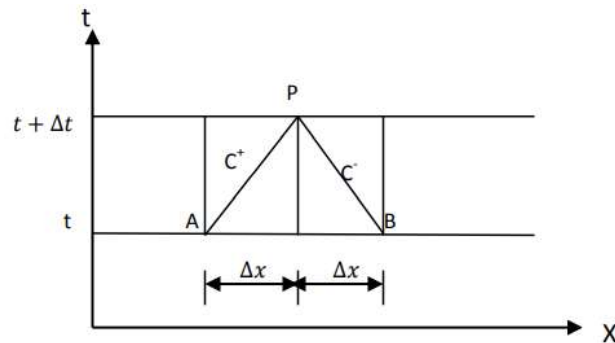
Equations (V) and (VI) are the characteristics equations in ODE, which can be solved by FDM.

$V \frac{\partial V}{\partial x}$ in eq. (I) and $V \frac{\partial P}{\partial x}$ in eq. (II) are much smaller than $\frac{\partial V}{\partial t}$ and $\frac{\partial P}{\partial t}$. So, they can also be neglected and $dx/dt = \pm c$.

6.2.1 Finite difference solution of characteristics equations of unsteady pipe flow

Characteristics equations are converted to finite difference equations by integration.

A) Finite difference equation in terms of velocity and pressure (using fixed grid)



Given or known: Velocity and pressure at previous time levels, i.e. V_A , P_A , x_A , t_A at A and V_B , P_B , x_B , t_B at B .

Slope of line $AP = dt/dx = 1/c$ and Slope of line $BP = dt/dx = -1/c$

Point of intersection of AP and BP represents the solution at P .

Taking C^+ (Line PA)

$$\frac{dV}{dt} + \frac{1}{\rho c} \frac{dP}{dt} + g \sin \theta + \frac{f V |V|}{2D} = 0$$

Multiplying throughout by $\rho c dt$ and integrating from A to P

$$\rho c \int_A^P dV + \int_A^P dP + \int_A^P \rho c g \sin \theta dt + \int_A^P \rho c dt \frac{f V |V|}{2D} = 0$$

As $dt/dx = 1/c$, replace cdt by dx .

$$\rho c \int_A^P dV + \int_A^P dP + \int_A^P \rho g \sin \theta dx + \int_A^P \rho dx \frac{fV|V|}{2D} = 0$$

Writing in Finite difference form

$$\rho c(V_P - V_A) + (P_P - P_A) + \rho g \sin \theta \Delta x + \frac{\rho \Delta x f V_A |V_A|}{2D} = 0 \quad (\text{VII})$$

Similarly, for C- (Line PB)

$$\frac{dV}{dt} - \frac{1}{\rho c} \frac{dP}{dt} + g \sin \theta + \frac{f V |V|}{2D} = 0$$

$$\text{i.e., } \rho c \int_B^P dV - \int_B^P dP + \int_B^P \rho c g \sin \theta dt + \int_B^P \rho c dt \frac{fV|V|}{2D} = 0$$

$$\text{i.e., } \rho c(V_P - V_B) - (P_P - P_B) + \rho g \sin \theta \Delta x + \frac{\rho \Delta x f V_B |V_B|}{2D} = 0 \quad (\text{VIII})$$

$$(\text{cdt} = -dx \text{ and } \int_B^P -dx = -(x_P - x_B) = -(-\Delta x) = \Delta x)$$

Eqs. (VII) and (VIII) are solved simultaneously to obtain V_P and P_P .

B) Finite difference equations in terms of hydraulic gradient line (H) and discharge (Q)

$$P = \rho g(H - Z)$$

Applying this equation at P, A and B

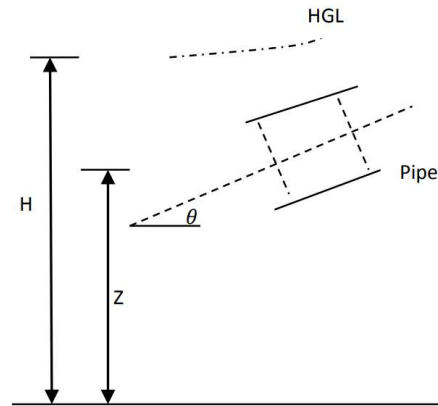
$$P_P = \rho g(H_P - Z_P)$$

$$P_A = \rho g(H_A - Z_A)$$

$$P_B = \rho g(H_B - Z_B)$$

Subtracting

$$\begin{aligned} P_P - P_A &= \rho g(H_P - Z_P) - \rho g(H_A - Z_A) = \rho g(H_P - H_A) - \rho g(Z_P - Z_A) \\ &= \rho g(H_P - H_A) - \rho g \Delta Z = \rho g(H_P - H_A) - \rho g \sin \theta \Delta x \end{aligned}$$



$$P_P - P_B = \rho g(H_P - Z_P) - \rho g(H_B - Z_B) = \rho g(H_P - H_B) - \rho g(Z_P - Z_B) \\ = \rho g(H_P - H_B) - \rho g dZ = \rho g(H_P - H_B) - \rho g \sin \theta \Delta x$$

Substituting value of $P_P - P_A$ in eq. (VII) and replacing V by Q/A [For C^+]

$$\frac{\rho c}{A}(Q_P - Q_A) + \rho g(H_P - H_A) - \rho g \sin \theta \Delta x + \rho g \sin \theta \Delta x + \frac{\rho \Delta x f Q_A |Q_A|}{2DA^2} = 0$$

Dividing throughout by ρg

$$\frac{c}{gA}(Q_P - Q_A) + (H_P - H_A) + \frac{f \Delta x Q_A |Q_A|}{2gDA^2} = 0$$

$$\text{i.e., } H_P = H_A - \frac{c}{gA}(Q_P - Q_A) - \frac{f \Delta x Q_A |Q_A|}{2gDA^2}$$

$$\text{i.e., } H_P = H_A - B(Q_P - Q_A) - R Q_A |Q_A| \quad (\text{IX})$$

$$\text{Taking } B = \frac{c}{gA} \text{ and } R = \frac{f \Delta x}{2gDA^2}$$

Substituting value of $P_P - P_B$ in eq. (VIII) and replacing V by Q/A [For C^-]

$$\frac{\rho c}{A}(Q_P - Q_B) - \rho g(H_P - H_B) - \rho g \sin \theta \Delta x + \rho g \sin \theta \Delta x + \frac{\rho \Delta x f Q_B |Q_B|}{2DA^2} = 0$$

Dividing throughout by ρg

$$\frac{c}{gA}(Q_P - Q_B) - (H_P - H_B) + \frac{f \Delta x Q_B |Q_B|}{2gDA^2} = 0$$

$$\text{i.e., } H_P = H_B + \frac{c}{gA}(Q_P - Q_B) + \frac{f \Delta x Q_B |Q_B|}{2gDA^2}$$

$$\text{i.e., } H_P = H_B + B(Q_P - Q_B) + R Q_B |Q_B| \quad (\text{X})$$

$$\text{Taking } B = \frac{c}{gA} \text{ and } R = \frac{f \Delta x}{2gDA^2}$$

C) Algorithm for solving water hammer problem (using rectangular grid, characteristics passing through diagonal)

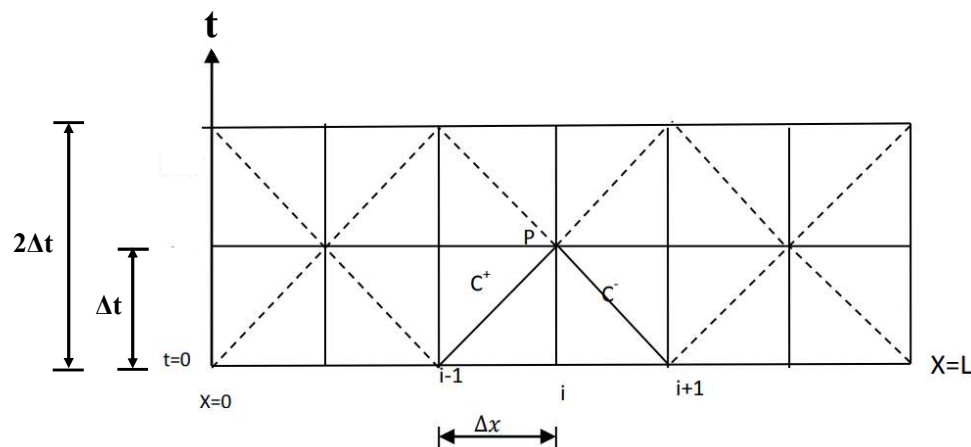
We have,

$$\text{For } C^+: H_P = H_A - B(Q_P - Q_A) - RQ_A|Q_A| \quad (a)$$

$$\text{For } C^-: H_P = H_B + B(Q_P - Q_B) + RQ_B|Q_B| \quad (b)$$

where $B = c/(gA)$, $R = f\Delta x/(2gDA^2)$, c = celerity of wave, A = c/s of pipe, f = friction factor, D = diameter of pipe.

Above equations are applied to compute values of H and Q at grid point i using rectangular grid.



Conditions at time $t = 0$ (i.e., initial condition must be given).

Take $\Delta x = L/N$ and $\Delta t = \Delta x/c$

where L = Length of pipe, and N = No. of reaches

Construct the rectangular grids in such a way that C^+ and C^- are the diagonals of the grid.

This approach is suitable for single pipe line.

Known: Q_{i-1} , H_{i-1} , Q_{i+1} , H_{i+1} (from initial condition or from previous time steps's computation)

To compute: H_{Pi} , Q_{Pi}

From eq. (a) (replacing A by $i-1$)

$$H_{Pi} = H_{i-1} - B(Q_{Pi} - Q_{i-1}) - RQ_{i-1} |Q_{i-1}|$$

$$\text{i.e., } H_{Pi} + BQ_{Pi} = H_{i-1} + Q_{i-1} (B - R|Q_{i-1}|)$$

$$\text{i.e., } H_{Pi} + BQ_{Pi} = C_p \quad (c) \quad \text{along } C^+$$

where $C_p = H_{i-1} + Q_{i-1} (B - R|Q_{i-1}|) = \text{known terms}$

From eq. (b) (replacing B by $i+1$)

$$H_{Pi} = H_{i+1} + B(Q_{Pi} - Q_{i+1}) + RQ_{i+1} |Q_{i+1}|$$

$$\text{i.e., } H_{Pi} - BQ_{Pi} = H_{i+1} - Q_{i+1} (B - R|Q_{i+1}|)$$

$$\text{i.e., } H_{Pi} - BQ_{Pi} = C_M \quad (d) \quad \text{along } C^-$$

where $C_M = H_{i+1} - Q_{i+1} (B - R|Q_{i+1}|) = \text{known terms}$

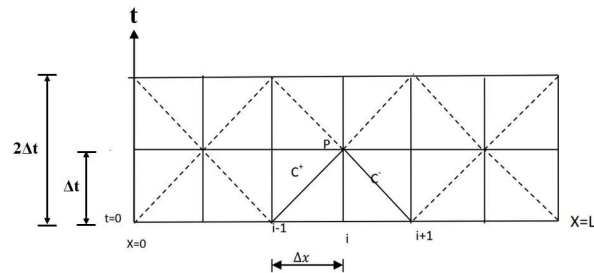
Solving equations (c) and (d)

$$H_{Pi} = (C_p + C_M)/2 \quad (e)$$

$$Q_{Pi} = (C_p - C_M)/2 \quad (f)$$

Algorithm

- Take $\Delta x = L/N$ and $\Delta t = \Delta x/c$.
- Compute B and R . [$B = c/(gA)$, and $R = f\Delta x/(2gDA^2)$]
- For time t
 - For grid point i , compute C_P and C_M .
 - Compute H_{Pi} and Q_{Pi} .
 - Perform similar computations for all interior nodes.
- Advance to next time step.



Stability of MOC solution

For stability of MOC solution, Courant number (C_n) should be less than 1.

$$C_n = \frac{|V + c|}{\Delta x / \Delta t}$$

For stability

$$C_n \leq 1$$

$$\text{i.e., } \frac{|V + c|}{\Delta x / \Delta t} \leq 1$$

$$\text{i.e., } \Delta t \leq \Delta x / |V + c|$$

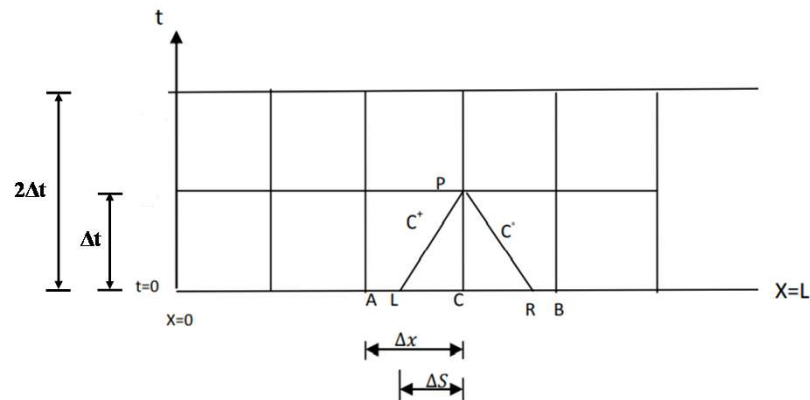
Divide pipes into N segments and compute Δt for all segments.

$$\text{i.e., } \Delta t \leq L / (N |V + c|)$$

Take minimum value of Δt for stability.

Solution technique for water hammer problem using specified time intervals (characteristics not on diagonal, interpolating values, time intervals known/given)

This approach is applicable for more than one pipes, where the characteristics do not meet the grid points.



Writing equations (a) and (b) by replacing A with L and B with R

$$\text{For } C^+: H_P = H_L - B(Q_P - Q_L) - RQ_L |Q_L| \quad (a)$$

$$\text{For } C^-: H_P = H_R + B(Q_P - Q_R) + RQ_R |Q_R| \quad (b)$$

Known value of H and Q at A , C and B at time $t = t_0$

Time interval: specified

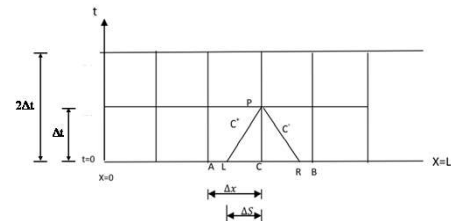
Co-ordinates of P , t_p , t_L , t_R : known.

Obtain H and Q at L and R by interpolation

$$\text{Coordinate of } L: X_L = X_P - \Delta S = X_P - c \Delta t = X_P - c(t_p - t_L)$$

$$\text{Coordinate of } R: X_R = X_P + \Delta S = X_P + c \Delta t = X_P + c(t_p - t_R)$$

$$\text{Linear interpolation} \quad (Q_L - Q_C)/(Q_A - Q_C) = \Delta S/\Delta X$$



$$Q_L = Q_C + (Q_A - Q_C) \Delta S / \Delta X \quad (I)$$

$$\Delta S / \Delta t = V_L + c \quad (\text{from } dx/dt = V + c)$$

As V_L is much smaller than c , neglect it.

$$\text{Then, } \Delta S / \Delta t = c \quad (II)$$

Substituting the value of $\Delta S / \Delta X$ from eq. (II) in eq. (I)

$$Q_L = Q_C + (Q_A - Q_C) c \Delta t / \Delta X \quad (III)$$

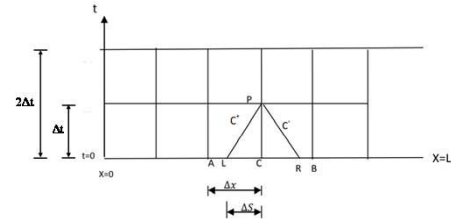
$$\text{Similarly, } Q_R = Q_C + (Q_B - Q_C) c \Delta t / \Delta X \quad (IV)$$

$$\text{Also, } (H_L - H_C) / (H_A - H_C) = \Delta S / \Delta X$$

$$\text{i.e., } H_L = H_C + (H_A - H_C) \Delta S / \Delta X$$

$$\text{i.e., } H_L = H_C + (H_A - H_C) c \Delta t / \Delta X \quad (V)$$

$$\text{i.e., } H_R = H_C + (H_B - H_C) c \Delta t / \Delta X \quad (VI)$$



6.3 Boundary conditions (BC) for unsteady pipe flow problem



At u/s end, only C^- line intersects the boundary and at d/s end, only C^+ line meets the boundary. As the intersection of two characteristics give the solution at a point for H and Q , an auxiliary equation is needed at each boundary. This is known as boundary conditions. This may be the end condition of the pipeline, or it may be a different element or facility attached to the end of the pipe.

The equations of boundary conditions used are: constant discharge or head, time varying discharge or head, relationship of discharge and head in algebraic or differential equation form.

Different types of boundaries in unsteady pipe flow problems

A. Reservoir at u/s end

HGL = constant for short duration

$H_{P1} = H_R$ = elevation of reservoir surface above datum

Reservoir level changing in known manner for longer times

e.g., $H_{P1} = H_R + \Delta H \sin \omega t$,

Q_{P1} obtained from eq. (f)

B. Discharge as a specified function of time at u/s end

$Q_{P1} = Q_0 + \Delta Q \sin \omega t$: known as BC

H_{P1} obtained from eq. (e)

C. Centrifugal pump at u/s end with head-discharge curve (table or equation) known

Pump supplying from a suction reservoir, whose surface elevation used as reference for HGL

$$H_{P1} = H_S + Q_{P1} (a_1 + a_2 Q_{P1})$$

H_S = shut off head, a_1 and a_2 = const. of curve

Solve this equation with eq. (f) simultaneously at the boundary.

D. Dead end at d/s end of pipe

Closed end at d/s (NS): $(Q_P)_{NS} = 0$

Obtain $(H_P)_{NS}$ from eq. (a or e).

E. Valve at d/s end of pipe

Datum for HGL taken at the valve

Orifice equation for steady flow

$$Q_O = (C_D A_G)_O \sqrt{2gH_O} \quad (I)$$

where Q_O is the steady state flow; H_O is the steady state head loss across the valve; and $(C_D A_G)_O$ is the discharge coefficient $\times$ area of valve opening.

For another opening

$$Q_P = C_D A_G \sqrt{2g \Delta H} \quad (II)$$

where ΔH is the instantaneous drop in HGL across the valve.

Let $\tau = C_D A_G / (C_D A_G)_O$

$$\text{Then } Q_P = \tau Q_O \sqrt{(\Delta H / H_O)} \quad (III)$$

For steady flow $\tau = 1$, and for closed valve with no flow $\tau = 0$.

Solve eq. (e) and (III) simultaneously.

F. Orifice at d/s end of pipe

Same equation as valve with $\tau = 1$

6.4 MOC for 1D gradually varied unsteady open channel flow

Saint Venant equations in terms of V and y without lateral flow are given as

Momentum equation:

$$\frac{\partial V}{\partial t} + V \frac{\partial V}{\partial x} + g \frac{\partial y}{\partial x} - g(S_0 - S_f) = 0 \quad (L_1)$$

Continuity equation:

$$\frac{\partial y}{\partial t} + V \frac{\partial y}{\partial x} + D \frac{\partial V}{\partial x} = 0 \quad (L_2)$$

Combining two equations as $L = L_1 + \lambda L_2 = 0$, where λ is the multiplier.

$$L = \left[\frac{\partial V}{\partial t} + (V + \lambda D) \frac{\partial V}{\partial x} \right] + \lambda \left[\frac{\partial y}{\partial t} + \left(V + \frac{g}{\lambda} \right) \frac{\partial y}{\partial x} \right] = g(S_0 - S_f) \quad (I)$$

We have

$$\frac{dV}{dt} = \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} \frac{dx}{dt} \text{ and } \frac{dy}{dt} = \frac{\partial y}{\partial t} + \frac{\partial y}{\partial x} \frac{dx}{dt}$$

Substituting in equation (I)

$$\frac{dV}{dt} + \lambda \frac{dy}{dt} = g(S_0 - S_f) \quad (II)$$

The above expression is valid

If $dx/dt = V + \lambda D$ for velocity part, and $dx/dt = V + g/\lambda$ for depth part

Equating these,

$$V + \lambda D = V + g/\lambda$$

$$\text{i.e., } \lambda = \pm \sqrt{(g/D)}$$

We have, celerity $c = \sqrt{gD}$ (Celerity is the function of flow depth, not constant like pipe flow)

Substituting $D = c^2 / g$

$$\lambda = \pm g/c$$

$$\text{i.e., } \frac{dV}{dt} \pm \frac{g}{c} \frac{dy}{dt} = g(S_0 - S_f) \quad (III)$$

With $\lambda = \pm g/c$,

$$dx/dt = V \pm c \quad (IV)$$

Equation (III) and (IV) are characteristic form of equations for 1D unsteady open channel flow.

Alternative expression for equation III

Considering rectangular channel: $c = \sqrt{g y}$

$$\text{i.e., } c^2 = g y$$

Differentiating w.r.t. t

$$2c \, dc/dt = g \, dy/dt$$

Substituting the value of $g \, dy/dt$ in equation (III)

$$\frac{dV}{dt} \pm 2 \frac{dc}{dt} = g(S_0 - S_f)$$

$$\text{i.e., } \frac{d(V \pm 2c)}{dt} = g(S_0 - S_f)$$

For equations in ODE

$$\text{For } +\text{ve characteristic: } \frac{d(V+2c)}{dt} = g(S_0 - S_f), \frac{dx}{dt} = V + c$$

For -ve characteristic: $\frac{d(V-2c)}{dt} = g(S_0 - S_f)$, $\frac{dx}{dt} = V - c$

For a horizontal frictionless channel or uniform flow ($S_0 = S_f = 0$)

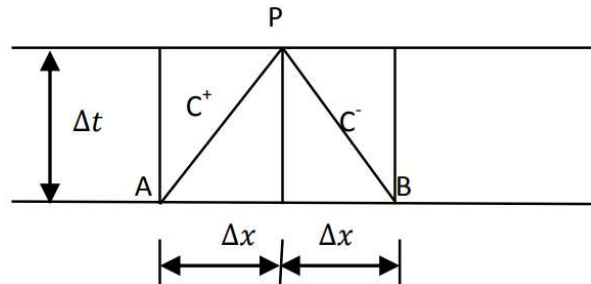
$V+2c = \text{constant} = J^+$ and $V-2c = \text{constant} = J^-$

$dx/dt = V+c$ and $dx/dt = V-c$

J^+ and J^- are called Riemann invariants.

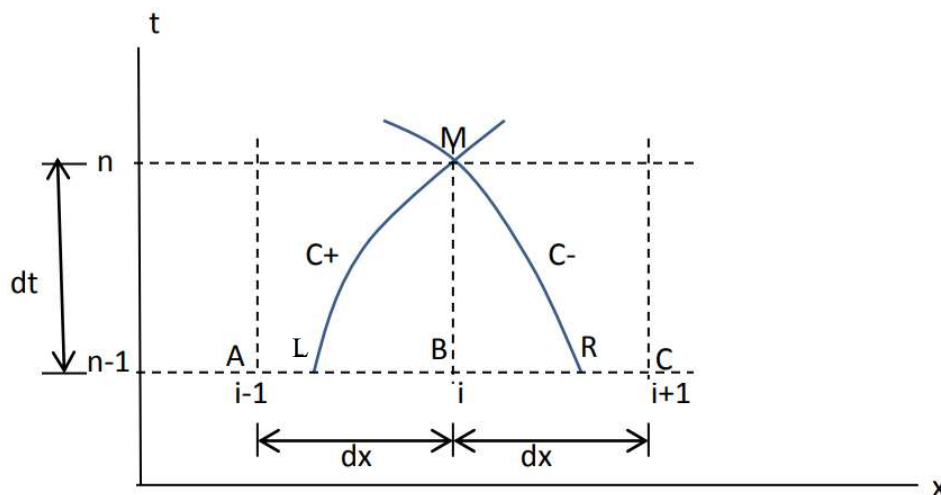
For C^+ : $V_A + 2C_A = V_P + 2C_P$

For C^- : $V_B - 2C_B = V_P - 2C_P$



Numerical (FDM) solution of MOC for 1D unsteady open channel flow

Fixed grid (Hartree method): fixed time and space interval

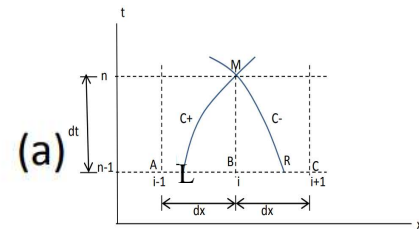


- Flow properties known at $t-1$: Depth (y) and velocity (V) known at L and R (As L and R are not on grid, velocity and celerity at these points should be found by interpolation from the known values at A , B and C).
- y and V required at M : intersection of C^+ and C^- at M gives the value of y and V at M .

Discretization of four ODES (characteristics form)

$$\frac{d(V+2c)}{dt} = g(S_0 - S_f)$$

$$\frac{(V_M+2C_M)-(V_L+2C_L)}{\Delta t} = g(S_0 - S_f)$$



$$\frac{d(V-2c)}{dt} = g(S_0 - S_f)$$

$$\frac{(V_M-2C_M)-(V_R-2C_R)}{\Delta t} = g(S_0 - S_f) \quad (b)$$

$$\frac{dx}{dt} = V + c$$

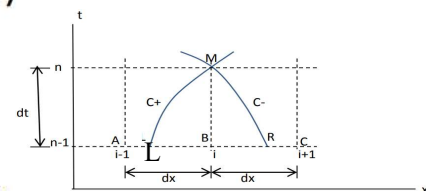
$$\frac{X_M - X_L}{\Delta t} = V_L + C_L$$

(c)

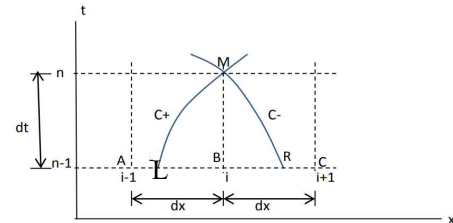
$$\frac{dx}{dt} = V - c$$

$$\frac{X_M - X_R}{\Delta t} = V_R - C_R$$

(d)



Unknowns in above equations (V_M , C_M , X_L , X_R) can be solved algebraically. X_L and X_R do not coincide with grid points. As the grid is fixed, X_M is known.

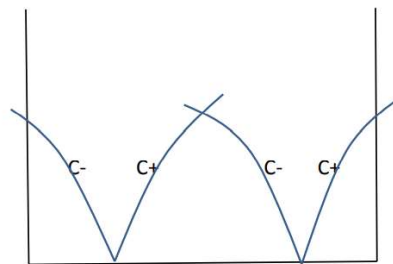


Requirements for solution

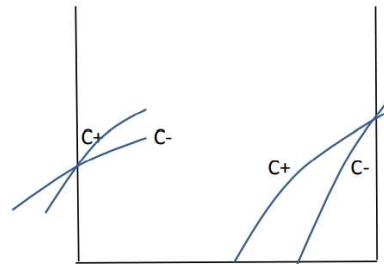
- Initial condition: V and y at $t = 0$
- Boundary conditions (BC): conditions at the boundary of physical system

Subcritical flow: both u/s and d/s BC

Supercritical flow: only u/s BC (2 conditions)



Subcritical
(one Characteristics intersecting at boundary)



Supercritical
(two Characteristics intersecting at boundary)

Types of BC

- water depth (y) or the free-surface elevation
- flow rate $Q(t)$ or flow velocity $V(t)$
- Rating curve (relationship between the water depth and flow)